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|                                                                                                                                                                                                                                                                                                               | Volume 51, Issue 2, July 2011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ISSN 0191-8869                              |
| <b>PERSONALITY AND<br/>INDIVIDUAL DIFFERENCES</b>                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                             |
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| OFFICIAL JOURNAL OF THE INTERNATIONAL SOCIETY FOR THE STUDY OF INDIVIDUAL<br>DIFFERENCES (ISSID)                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                             |

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# Personality and Individual Differences

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## On the stability of instability: Optimising the reliability of intra-subject variability of reaction times

Christopher W.N. Saville<sup>a,b</sup>, Ralph Pawling<sup>a</sup>, Mark Trullinger<sup>a</sup>, David Daley<sup>c</sup>, James Intriligator<sup>a</sup>, Christoph Klein<sup>a,b,\*</sup>

<sup>a</sup> School of Psychology, Bangor University, United Kingdom

<sup>b</sup> University Hospital for Child and Adolescent Psychiatry, University of Freiburg, Germany

<sup>c</sup> Division of Child and Adolescent Psychiatry, School of Community Health Sciences, University of Nottingham, United Kingdom

### ARTICLE INFO

#### Article history:

Received 28 October 2010

Received in revised form 16 March 2011

Accepted 21 March 2011

Available online 13 April 2011

#### Keywords:

Reliability

Reaction time variability

Ex-Gaussian

LATER

Reaction times

### ABSTRACT

While reaction times have been traditionally aggregated using measures of central tendency, interest in higher moments of the reaction time distribution, particularly intra-subject variability (ISV), has grown in recent years. However it is unclear to what extent individual differences in these higher moments are stable across time, reflecting trait-like features. The present study compares the reliability of a number of metrics for higher moments of the reaction time distribution on a battery of speeded tasks, and supplements these analyses with a Monte Carlo analysis of the go no-go task. The reliability of ISV is shown to be dependent on both the metric used and the number of trials used to calculate them. However, when using sufficient trials and appropriate metrics, ISV shows good test–retest and odd–even reliability. This study has important practical implications for the design and analysis of studies into ISV, as well as the theoretical importance for the trait concept of ISV.

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### 1. Introduction

Experimental psychology has a long tradition of using reaction times (RTs), in order to identify the mechanisms underpinning mental operations. In order to obtain reliable measurements, cognitive experiments involve many trials so require summary statistics to make sense of the data. Generally a measure of central tendency, such as the mean or median, is used to characterise the RT distribution, providing an easily interpretable measure of a participant's RT. However, RTs come from multi-parameter distributions and the use of means and medians to the exclusion of other measures distract researchers from seeing that variation around the mean may be more than just noise, and instead may be another facet of the signal.

While the study of higher moments of the RT distribution has yet to find widespread favour in mainstream experimental psychology, the topic is rapidly gaining importance within differential psychology and neuroscience. Intra-subject variability (ISV) of reaction times, reflecting the intra-individual variation of response times around the mean and thus the second moment of the RT distribution, has been shown to exhibit strong relationships with a

number of neurological and neuropsychiatric conditions, including attention deficit hyperactivity disorder (ADHD, [Klein, Wendling, Huettner, Ruder, & Peper, 2006](#)), schizophrenia (Birkett et al., 2007), Alzheimer's (Burton, Strauss, Hultsch, Moll, & Hunter, 2006) as well as old age (Hultsch, MacDonald, & Dixon, 2002) and terminal decline (MacDonald, Hultsch, & Dixon, 2008). It is also related to psychometric intelligence, with correlations between ISV and intelligence surpassing those between mean RT (MRT) and intelligence ([Jensen, 1992](#)). Investigating the higher moments of RT distributions is, however, not without complications. A great many measures of ISV exist, and there are few studies comparing them. There is also the issue of reliability, which is of cardinal importance in differential psychology.

#### 1.1. Choice of parameter

Most ISV research uses measures of variance, such as standard deviation (SD) or mean absolute deviation (MAD). These parameters measure the spread of observations around the mean, irrespective of the direction of the deviation. RT distributions, however, are not Gaussian and show significant rightward skew. It is thus questionable whether such measures of ISV are appropriate, or whether they make an implicit but erroneous assumption of Gaussianity.

Indeed there is evidence suggesting that this asymmetry, characterised by an extended tail at the slow end of the RT distribution,

\* Corresponding author. Address: School of Psychology, Bangor University, Brigantia Building, Penrallt Road, Bangor, Gwynedd LL57 2AS, United Kingdom. Tel.: +44 1248 388 351; fax: +44 1248 382 599.

E-mail address: [c.klein@bangor.ac.uk](mailto:c.klein@bangor.ac.uk) (C. Klein).

may be important. [Larson and Alderton \(1990\)](#) coined the *worst performance rule* by showing that slow RTs are more highly correlated with IQ than fast RTs. These right-tail RTs also discriminate between children with and without ADHD ([Leth-Steensen, King-Elbaz, & Douglas, 2000](#)), and thus may be of special interest to differential psychologists.

Several parameters have been suggested to quantify the right tail of the distribution. Skewness is the third moment of a distribution, and captures the distribution's asymmetry. As a higher moment however, it may require a large number of observations to be reliably estimated, so most researchers use other approaches. The  $\tau$  parameter of the ex-Gaussian distribution is another important measure of skewness. The ex-Gaussian is a convolution of a Gaussian and an exponential distribution, with the parameters  $\mu$  and  $\sigma$  respectively representing the mean and standard deviation of the Gaussian component, and  $\tau$  representing the mean and standard deviation of the exponential component, which gives the ex-Gaussian its positive skew.

While ex-Gaussian parameters represent one approach for characterising RT distributions, others exist. [Carpenter \(1981\)](#) advocates carrying out analyses on the reciprocals of RTs, rather than on raw values. By taking the reciprocal of an RT, we treat it as a rate rather than a latency. Reciprocal RT distributions appear Gaussian, and do not show the skewness that characterises raw RTs ([Carpenter, 1981](#)). The first two moments of this distribution can be measured by taking the mean and standard deviations of the reciprocals of the RTs, or by fitting Carpenter's Linear Approach to Threshold with Ergodic Rate (LATER) model, which gives the mean and standard deviation of the best-fitting Gaussian distribution of the reciprocal RTs. This model, however, has as yet largely been used in saccadic RT research only.

It remains unclear whether these different parameters measure separate aspects of ISV, or whether they render one another largely redundant (but see [Schmiedek, Oberauer, Wilhelm, Süß, & Wittmann, 2007](#)). This raises problems when comparing across studies that use different measures of ISV. On one hand, we may be glossing over genuine differences in what various parameters measure; on the other, we may be overemphasising differences between metrics that essentially measure the same thing.

### 1.2. Reliability of ISV

A second key issue for the use of ISV for individual differences research is reliability. According to test theory ([Novick, 1966](#)), all measurements represent a true score, plus error variance. The reliability of a measurement is the extent to which it measures the true score, rather than error. ISV was long thought to represent error variance, rather than a trait construct, and it is in part the apparent reliability of ISV ([Flehmig, Steinborn, Langner, Scholz, & Westhoff, 2007](#); [Kuntsi, Stevenson, Oosterlan, & Sonuga-Barke, 2001](#)) which led to its recognition as a variable of interest. ISV does, however, appear to be less reliable than measures of central tendency ([Jensen, 1992](#)) raising the concern that reliability may be an issue for ISV. This is particularly important as ISV is a candidate marker for ADHD, schizophrenia, cognitive ageing, intelligence, and terminal decline. ISV is clearly only of use as a marker if it exhibits adequate reliability.

Although there is work suggesting that ISV may be reliable, albeit less so than measures of central tendency, this work is restricted to a small number of metrics. It remains unknown how other measures, such as ex-Gaussian and LATER parameters, compare to more common measures like standard deviations and coefficients of variation.

One determining factor of the reliability of ISV is the number of trials it is based upon. According to the Spearman–Brown formula ([Brown, 1910](#); [Spearman, 1910](#)), the Gaussian distribution of error

variance leads to error cancelling itself out with increasing aggregation of data, leaving the true score intact.

Such a conceptualisation of data aggregation is however strongly tied to the idea of a true score being analogous to a measure of central tendency of the distribution. Estimating the variance of a distribution may show a less transparent relationship between data aggregation and reliability. Measures of ISV could need greater numbers of RTs to 'settle' than first moments such as means and medians, and be more vulnerable to outlier observations. Furthermore, ISV, by definition, must be estimated from a number of RTs, whereas average RT could, assuming perfect reliability, be measured from a single RT. This implies that the relationship between number of trials and ISV may go beyond simple data aggregation. Despite this, at present many researchers use similar numbers of trials for estimating ISV as they would for estimating measures of central tendency, a practice which may yield unreliable measures of ISV.

The aims of the present study are therefore threefold. Firstly, we set out to derive a large set of different ISV metrics from standard neuropsychological tasks that are popular in clinical research and explore the relationships between these measures. Are different measures of ISV redundant with regard to each other, or do they offer different information? Secondly, we determined the instrumental (odd–even) and test–retest reliabilities of these measures to assess their appropriateness as trait correlates in differential research. Which measures of ISV are most reliable? Finally, we conducted a Monte Carlo analysis of the test–retest reliability of the mean and standard deviation of RTs with different numbers of trials. Is the increase in reliability that the Spearman–Brown formula predicts with increasing trials slower for ISV than mean RT?

## 2. Methodology

### 2.1. Participants

Eighty-seven student volunteers from Bangor University participated in exchange for printer credits and course credit. Twelve participants' data were excluded due to non-completion of the second testing session and a further ten participants' data were excluded due to responding correctly on fewer than 30% of trials on any condition on any of the six tasks (list-wise exclusion). This left  $N = 65$  participants which all reported analyses are based upon. Participants who showed consistently high or low ISV were recruited back for a separate electroencephalographic study; interested readers are referred to [Saville et al. \(in press\)](#).

### 2.2. Apparatus

Participants were tested on Pentium IV personal computers running Windows. Stimuli were presented on liquid crystal display monitors measuring  $382 \times 300$  mm and refreshing at 60 Hz, and participants responded using Qwerty keyboards.

### 2.3. Design

The study used a within-subject test–retest correlational design.

### 2.4. Procedure

Participants were tested on two occasions exactly one week apart. On both occasions they carried out the six tasks, described below, used by [Klein et al. \(2006\)](#). Each session was held in the same location, at the same time of day, and was overseen by the same experimenter in order to maximise consistency between

testing sessions. Task order was counterbalanced between participants using a Latin square design, but was kept constant across sessions for each participant.

In the *0-back task* participants watched a series of letters (100 trials) appear on the screen. They were instructed to press the F key every time a letter appeared, unless it was the letter E (occurring in 20% of trials), in which case they were to press the J key. The *1-back task* was identical to the 0-back task, except that instead of pressing J when an E appeared, they pressed J whenever a letter matched the preceding letter. The *2-back task* was identical to the above tasks, but the J key was to be pressed when a letter matched the last but one letter. Henceforth, trials requiring an F response will be termed *non-events* and those requiring a J response will be termed *events*. These tasks will also be referred to collectively as *n-back tasks*.

In the continuous performance task, participants watched a series of letters (300 trials) appear on the screen and pressed the space key whenever an X appeared (46 trials). To all other letters they were instructed to withhold any response. The go-no go task was the opposite of the continuous performance task; participants were asked to respond to every letter, only withholding responses to the letter X (thus there were 254 trials requiring a response).

In the stop signal task, participants watched a series (300 trials) of Ls and Rs appear on the screen, and were asked to press the F key in response to Ls and the J key in response to Rs. In 75 trials, however, a red stop sign appeared and participants were instructed to withhold responses on trials when this happened. This left 225 trials where responses were required. The stop sign was presented 150 ms post-stimulus on the first stop trial, and this latency was increased by 25 ms every time the participant successfully withheld their response and was decreased by 25 ms every time they responded.

In all tasks the participants were asked to respond as quickly and accurately as possible to every letter.

## 2.5. Data analysis

In order to assess the reliability of ISV, several parameters were computed. Unless stated, all parameters were computed separately for each participant on each task, session, and, where appropriate, condition.

Firstly, an accuracy measure (correct responses/total number of trials) was computed. After this, trials where responses were either incorrect or were made within 120 ms (and thus were presumably pre-emptive) were removed from the data before RT parameters (see below) were computed.

The parameters fell into four groups, corresponding to the first four moments of a statistical distribution. The first group consisted of measures of central tendency: *Mean RT* (MRT), *median RT* (MnRT), the  $\mu$  parameter of the ex-Gaussian distribution, the mean reciprocal RT (RecipMRT), and the  $\mu$  parameter of the LATER model.

The second group was made up of measures of variance: *Standard deviation* (SDRT) is probably the most familiar measure of variability. Due to its squaring of deviations in order to obtain uniformly positive values, it gives a somewhat disproportionately high weighting to extreme scores. *Coefficient of variation* (CVRT), defined as SDRT/MRT, is a measure of how much variability exists per unit of MRT. *Mean absolute deviation* (MAD) is the mean deviation from the MRT, rectified so that opposite signed deviations do not cancel. As MAD uses absolute rather than squared values it should be less influenced by extreme values than SDRT. *Range* is simply the difference between the fastest and slowest RT. *Inter-quartile range* (IQR) is the difference between the first and third quartile of the RT distribution.  $\sigma$  is the estimated standard deviation of the Gaussian component of the ex-Gaussian distribution.

Finally the *standard deviation of reciprocal RTs* (RecipSDRT) and  $\sigma$  parameter of the LATER model were used.

The third group consisted of measures of asymmetry and of the size of the right-handed tail of the distribution: *skewness* is a measure of a distribution's asymmetry. RT distributions are generally positively skewed, possibly because RTs are effectively capped at the fast end, but largely uncapped in terms of how slow a response can be. Thus a high positive skew is indicative of these very slow responses.  $\tau$  is the estimated mean and standard deviation of the exponential component of the ex-Gaussian distribution.

The final measure was *kurtosis*, a measure of peakedness of a distribution, which indexes the extent to which the variance results from infrequent outliers, as opposed to more frequent smaller deviations from the average.

The three ex-Gaussian parameters were computed using quantile maximum probability estimation (QMPE; Heathcote, Brown, & Cousineau, 2004) software, available freely from the Newcastle Cognition Lab website (<http://www.newcl.org/>). The authors suggest that meaningful parameters can be calculated from as few as 40 observations. The *n-back* tasks only had a maximum of 20 event trials however, and so ex-Gaussian parameters were not calculated for these trials.

The LATER model was fitted using SPIC, a freely available software package (Carpenter, 1994; <http://www.cudos.ac.uk/spic.htm>). As it is inadvisable to fit the LATER model using fewer than 100 trials (Carpenter, personal communication), the two LATER parameters were only derived for the go-no go task.

Pearson's *r* coefficients were computed between session one and two for each parameter in order to estimate test-retest reliability. Pearson's *r* coefficients were also derived for odd and even trials in order to measure instrumental reliability (for a discussion of these constructs in the context of RT research, see Klein & Fischer, 2005). Due to insufficient trials, odd-even reliability coefficients were not computed for *n-back* task event trials on any parameter, or for the *n-back* tasks or continuous performance task on the ex-Gaussian parameters. LATER parameters were only computed for the go-no go task.

In order to assess the extent to which different ISV parameters provided different information, correlation matrices were computed for the parameters from the go-no go task (the task with the most trials) across each occasion. The correlation coefficients were then averaged across sessions to obtain a single matrix.

While comparing the test-retest correlations for tasks with different numbers of trials gives some indication of the relationship between reliability and data aggregation, this is a rather indirect assessment of this relationship. Monte Carlo approaches represent a far more rigorous method to explore this question. We sampled random (with replacement) subsets of trials from each participant's performance on the go-no go task, separately for sessions one and two, derived MRTs and SDRTs from these trials and computed test-retest correlation coefficients. This was carried out 100 times for each number of trials, which ranged from 10 to 300 in 10 trial increments. Random with replacement sampling was used instead of random without replacement sampling, in order to obtain meaningful estimates of variance in reliability coefficients.

## 3. Results

Test-retest reliability coefficients for the various tasks and parameters can be seen in Table 1. The data show that the event trials of the *n-back* tasks exhibited the lowest levels of reliability, particularly for the 0-back task. This is likely due to the small number of these trials (20). In contrast the go-no go task, which required 254 responses, shows good reliability across most measures of ISV.



**Table 1**

Test–retest correlation coefficients for RT parameters on each task, and median number of trials (averaged across sessions) used to compute them (OBT = 0-back task, 1BT = 1-back task, 2BT = 2-back task, CPT = continuous performance task, GNG = go-no go task, SST = stop signal task, *e* = event trials, *ne* = non-event trials).

|                   | Median Trials used | MRT   | SDRT  | CVRT  | MAD   | Skew   | Kurt   | MnRT  | Range | IQR   | RecipMRT | RecipSDRT | Ex-G $\mu$ | Ex-G $\sigma$ | Ex-G $\tau$ | LATER $\mu$ | LATER $\sigma$ |
|-------------------|--------------------|-------|-------|-------|-------|--------|--------|-------|-------|-------|----------|-----------|------------|---------------|-------------|-------------|----------------|
| OBT <sub>e</sub>  | 18                 | 0.614 | 0.213 | 0.125 | 0.241 | 0.140  | 0.075  | 0.592 | 0.197 | 0.134 | 0.704    | 0.374     | –          | –             | –           | –           | –              |
| 1BT <sub>e</sub>  | 17                 | 0.796 | 0.567 | 0.543 | 0.495 | 0.390  | 0.273  | 0.794 | 0.604 | 0.377 | 0.830    | 0.594     | –          | –             | –           | –           | –              |
| 2BT <sub>e</sub>  | 13.5               | 0.851 | 0.604 | 0.269 | 0.620 | –0.205 | –0.198 | 0.798 | 0.518 | 0.527 | 0.743    | 0.076     | –          | –             | –           | –           | –              |
| OBT <sub>ne</sub> | 79                 | 0.645 | 0.565 | 0.391 | 0.681 | 0.034  | –0.075 | 0.641 | 0.289 | 0.741 | 0.750    | 0.497     | 0.631      | 0.608         | 0.486       | –           | –              |
| 1BT <sub>ne</sub> | 78                 | 0.811 | 0.603 | 0.471 | 0.639 | 0.162  | 0.233  | 0.836 | 0.482 | 0.614 | 0.849    | 0.511     | 0.849      | 0.518         | 0.551       | –           | –              |
| 2BT <sub>ne</sub> | 71                 | 0.832 | 0.812 | 0.506 | 0.821 | 0.487  | 0.360  | 0.800 | 0.637 | 0.766 | 0.835    | 0.544     | 0.681      | 0.521         | 0.118       | –           | –              |
| CPT               | 46                 | 0.665 | 0.609 | 0.431 | 0.645 | 0.268  | 0.178  | 0.653 | 0.480 | 0.519 | 0.783    | 0.389     | 0.569      | 0.310         | 0.384       | –           | –              |
| GNG               | 253                | 0.836 | 0.842 | 0.862 | 0.804 | 0.250  | 0.120  | 0.837 | 0.727 | 0.663 | 0.808    | 0.803     | 0.912      | 0.771         | 0.753       | 0.630       | 0.920          |
| SST               | 218.5              | 0.831 | 0.499 | 0.300 | 0.454 | 0.464  | 0.305  | 0.802 | 0.477 | 0.370 | 0.868    | 0.499     | 0.773      | 0.348         | 0.404       | –           | –              |

The measures of central tendency (MRT, MnRT, and the two  $\mu$  parameters) consistently outperformed measures of ISV on both test–retest (Table 1) and odd–even (Table 2) reliability, replicating previous findings (Jensen, 1992). SDRT, MAD, and LATER  $\sigma$  were the best performing measures of ISV (although the latter parameter was only available for the go-no go task) beating other common measures of ISV, such as CVRT,  $\tau$ , and IQR across most tasks. Skew and kurtosis showed poor reliability across the board and cannot be recommended for use in individual differences research.

Odd–even reliability coefficients are presented in Table 2. Most parameters show very good reliability, especially on the go-no go task – which had the most trials – where reliability coefficients are mainly .90 or above. Again SDRT, MAD and LATER  $\sigma$  performed well, and this time RecipSDRT and IQR performed similarly well.

Table 3 shows the correlation matrix of the sixteen parameters for the go-no go task, averaged across sessions. There are substantial correlations between most parameters, even between parameters that ostensibly measure different moments of the distribution. For example, ex-Gaussian  $\tau$ , essentially a measure of skewness, is almost perfectly correlated ( $r = .96$ ) with SDRT, a measure of variance. Some of the most reliable ISV parameters (SDRT, MAD) are so highly correlated that they are essentially redundant with regard to one another. Interestingly RecipSDRT and LATER  $\sigma$  appeared to be largely uncorrelated with MRT and MnRT, unlike other variability parameters which were all substantially correlated with measures of central tendency. MnRT appeared more independent of the measures of ISV than MRT, possibly due to its greater robustness to outliers.

The results of the Monte Carlo analysis are presented in Fig. 1. Here reliability on the Y-axis is graphed against number of trials on the X-axis for MRT and SDRT. While MRT begins to reach a plateau after only 50 trials, SDRT does not reach a comparable plateau

until as many as 200 trials, showing that SDRT needs a much greater level of data aggregation to achieve adequate reliability than MRT.

#### 4. Discussion

The present psychometric study of different metrics of ISV, assessed for six established neuropsychological tests with a sample of 65 undergraduate students, yielded the following main results. First, given sufficient levels of data aggregation (i.e. based on the go-no go task), most measures of variability exhibited good to excellent instrumental reliabilities, with odd–even correlations reaching as high as .95 for MAD, and around the .90 mark for many other parameters. Measures of the higher moments of the distribution, namely skewness and kurtosis, however exhibited poor levels of reliability. Second, these metrics also showed good test–retest reliabilities, reaching as high as .92 for LATER  $\sigma$ , with most measures surpassing .80. Thirdly, most metrics showed substantial inter-correlations, to the point where many metrics could be argued to be redundant with one another. This was true for metrics that are clearly largely statistically interdependent (e.g. SDRT and MAD), but, more surprisingly, also true for SDRT and  $\tau$ , metrics which theoretically measure different moments of the RT distribution.

The key finding of the present study is that metrics of ISV show good test–retest and odd–even reliability, and are thus suitable for individual differences research. These data do however draw attention to the importance of the number of trials used to compute such metrics. Our Monte Carlo simulation shows a striking difference between MRT and SDRT in the number of trials needed to achieve equivalent reliability. We thus encourage researchers utilising measures of ISV to consider using a greater number of trials

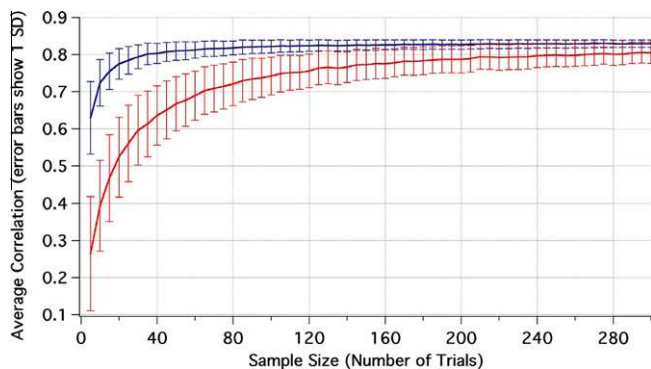
**Table 2**

Odd–even reliabilities for RT parameters on each task (OBT = 0-back task, 1BT = 1-back task, 2BT = 2-back task, CPT = continuous performance task, GNG = go-no go task, SST = stop signal task).

| Task | Session | MRT   | SDRT  | CVRT  | MAD   | Skew   | Kurt   | MnRT  | Range | IQR   | RecipMRT | RecipSDRT | Ex-G $\mu$ | Ex-G $\sigma$ | Ex-G $\tau$ | LATER $\mu$ | LATER $\sigma$ |
|------|---------|-------|-------|-------|-------|--------|--------|-------|-------|-------|----------|-----------|------------|---------------|-------------|-------------|----------------|
| OBT  | 1       | 0.986 | 0.805 | 0.698 | 0.872 | 0.123  | –0.057 | 0.980 | 0.534 | 0.816 | 0.977    | 0.783     |            |               |             |             |                |
|      | 2       | 0.989 | 0.702 | 0.584 | 0.837 | 0.180  | 0.030  | 0.983 | 0.470 | 0.776 | 0.981    | 0.690     |            |               |             |             |                |
| 1BT  | 1       | 0.982 | 0.503 | 0.344 | 0.672 | 0.151  | –0.028 | 0.983 | 0.308 | 0.746 | 0.982    | 0.723     |            |               |             |             |                |
|      | 2       | 0.989 | 0.773 | 0.659 | 0.859 | 0.278  | 0.216  | 0.984 | 0.554 | 0.838 | 0.985    | 0.739     |            |               |             |             |                |
| 2BT  | 1       | 0.977 | 0.840 | 0.696 | 0.866 | 0.506  | 0.264  | 0.963 | 0.645 | 0.810 | 0.975    | 0.741     |            |               |             |             |                |
|      | 2       | 0.772 | 0.476 | 0.292 | 0.552 | –0.045 | 0.033  | 0.767 | 0.326 | 0.529 | 0.796    | 0.445     |            |               |             |             |                |
| CPT  | 1       | 0.965 | 0.724 | 0.569 | 0.711 | 0.244  | 0.507  | 0.964 | 0.692 | 0.391 | 0.952    | 0.411     |            |               |             |             |                |
|      | 2       | 0.973 | 0.472 | 0.267 | 0.589 | 0.399  | 0.265  | 0.980 | 0.353 | 0.610 | 0.968    | 0.353     |            |               |             |             |                |
| GNG  | 1       | 0.995 | 0.911 | 0.866 | 0.950 | 0.340  | 0.239  | 0.995 | 0.735 | 0.902 | 0.994    | 0.907     | 0.987      | 0.857         | 0.903       | 0.986       | 0.897          |
|      | 2       | 0.990 | 0.940 | 0.920 | 0.959 | 0.442  | 0.419  | 0.995 | 0.802 | 0.961 | 0.985    | 0.897     | 0.974      | 0.784         | 0.944       | 0.990       | 0.947          |
| SST  | 1       | 0.996 | 0.946 | 0.833 | 0.965 | 0.599  | 0.270  | 0.995 | 0.781 | 0.948 | 0.992    | 0.919     | 0.980      | 0.936         | 0.747       |             |                |
|      | 2       | 0.998 | 0.941 | 0.852 | 0.970 | 0.500  | 0.221  | 0.997 | 0.690 | 0.955 | 0.996    | 0.949     | 0.986      | 0.916         | 0.694       |             |                |

**Table 3**  
Intercorrelations for RT parameters on the go-no go task.

|                | MRT   | SDRT  | CVRT  | MAD   | Skew  | Kurt  | MnRT  | Range | IQR   | Ex-G $\mu$ | Ex-G $\sigma$ | Ex-G $\tau$ | RecipMRT | RecipSDRT | LATER $\mu$ |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------------|---------------|-------------|----------|-----------|-------------|
| SDRT           | .713  |       |       |       |       |       |       |       |       |            |               |             |          |           |             |
| CVRT           | .383  | .915  |       |       |       |       |       |       |       |            |               |             |          |           |             |
| MAD            | .779  | .985  | .852  |       |       |       |       |       |       |            |               |             |          |           |             |
| Skew           | -.057 | .301  | .469  | .166  |       |       |       |       |       |            |               |             |          |           |             |
| Kurt           | -.086 | .149  | .288  | .025  | .934  |       |       |       |       |            |               |             |          |           |             |
| MnRT           | .989  | .612  | .259  | .689  | -.122 | -.116 |       |       |       |            |               |             |          |           |             |
| Range          | .607  | .917  | .885  | .865  | .572  | .472  | .515  |       |       |            |               |             |          |           |             |
| IQR            | .835  | .908  | .714  | .961  | .022  | -.075 | .771  | .775  |       |            |               |             |          |           |             |
| Ex-G $\mu$     | .881  | .319  | -.059 | .405  | -.236 | -.163 | .933  | .2485 | .507  |            |               |             |          |           |             |
| Ex-G $\sigma$  | .792  | .542  | .290  | .603  | -.213 | -.147 | .810  | .4735 | .656  | .772       |               |             |          |           |             |
| Ex-G $\tau$    | .643  | .962  | .892  | .960  | .270  | .086  | .537  | .858  | .901  | .218       | .386          |             |          |           |             |
| RecipMRT       | -.931 | -.659 | -.359 | -.720 | .008  | .061  | -.914 | -.579 | -.775 | -.799      | -.628         | -.637       |          |           |             |
| RecipSDRT      | -.001 | .639  | .857  | .573  | .315  | .176  | -.110 | .619  | .434  | -.382      | .172          | .615        | .070     |           |             |
| LATER $\mu$    | -.836 | -.706 | -.472 | -.744 | .002  | .065  | -.799 | -.587 | -.752 | -.656      | -.605         | -.655       | .872     | -.118     |             |
| LATER $\sigma$ | .003  | .586  | .778  | .518  | .236  | .100  | -.104 | .500  | .366  | -.339      | .110          | .550        | .088     | .881      | -.192       |



**Fig. 1.** Summary of results for Monte Carlo analysis of test-retest reliability on the go-no go task. Blue bar represents MRT and red bar represents SDRT. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

than a standard RT experiment, as these parameters require a greater number of trials than the measures of central tendency which psychologists are used to.

Another important upshot of the high test-retest reliability of ISV metrics observed in the more successful tasks of the present study is that it provides further evidence that ISV shows trait-like characteristics. This is important for the idea of ISV as a suitable topic for differential psychology, but also as a possible endophenotype for psychiatric disorders – a key topic in ISV research. These data also suggest that ISV is a relatively stable trait in healthy participants, rather than only being a feature of pathological states such as psychosis or terminal decline.

Our data also show that the majority of ISV metrics are largely redundant with regard to one another. Indeed, even measures that theoretically measure different moments of the RT distribution, *e.g.* SDRT and  $\tau$ , are highly intercorrelated. This has two important implications. Firstly, it may support the idea that the right tail of the distribution is of great theoretical importance. Tail RTs may be the source of much of the variance usually attributed to the second moment of a distribution, a notion supported by the stronger correlation between SDRT and  $\tau$  than SDRT and  $\sigma$ . Secondly, however, a pragmatic case may be made that even if the previous point is correct, the SDRT or MAD may still be the best choice of parameter for measuring ISV. In the light of the high intercorrelation between these measures, a researcher could do worse than to choose their parameter based on psychometric reliability, and many of the

most reliable measures, like SDRT and MAD, were relatively simple variance measures.

Despite ISV metrics being generally highly intercorrelated, it is worth mentioning the exceptions to the rule: measures based on reciprocals. RecipSDRT and LATER  $\sigma$  showed relatively low intercorrelations with other ISV measures. This may be, in part, due to their tendency to de-emphasise right-tail RTs compared to most ISV metrics. In light of the possible importance of these RTs, this may be undesirable for an individual differences researcher, but it is equally possible that they are measuring a different but equally interesting form of ISV. Future research should assess whether these metrics are less associated than traditional measures with constructs such as IQ and ADHD, which are thought to be related to the right tail of the RT distribution.

On the basis of the present data, we would advise researchers that the  $\sigma$  parameter from the LATER model, and either SDRT or MAD are good choices as measures of ISV. Despite the theoretical appeal of ex-Gaussian  $\tau$ , it is so highly correlated with SDRT that it may fail to add appreciable value to an analysis.

It is important to address some of the limitations of the present study. It is likely that the reliability of RT parameters is somewhat task-dependent, and, while the battery of tasks used includes some of the more common RT paradigms, it is not exhaustive. Furthermore, as some of our analyses were only possible for the go-no go task (LATER model, Monte-Carlo analysis, redundancy analysis), we are implicitly assuming that results obtained for this task can be generalised across tasks. For these reasons it is difficult to draw strong conclusions about which tasks are most appropriate for studying ISV. That said, our data do suggest that tasks which provide the greatest number of responses in the shortest period of time will yield superior reliability, and allow more sophisticated analyses, than tasks where fewer responses are made. The contrast between the reliabilities for the go-no go task, where participants responded to most trials, and the continuous performance task, where participants responded to only a minority of trials, illustrates this point well.

On this note, it is worth addressing the appropriateness of the stop signal task for ISV research. Due to the conflicting demands of the two subtasks (responding quickly to go trials while avoiding responding to stop trials), participants may employ strategies that prioritise one over the other. Not only does the fact that this task is not a 'pure' RT task itself raise concerns about using it for ISV research, but the strategy employed by a participant may vary from session to session, adversely affecting reliability. The relatively low test-retest, as compared to odd-even, reliability found in this task may be evidence of the latter problem in the present

study. Further research is needed to validate the use of this task for ISV research.

The disparity in number of trials between the different tasks also makes it difficult to compare the different tasks in terms of their reliability. While there is a relationship between the number of trials in a task and the reliability of that task, tasks with similar numbers of trials (such as the *n*-back tasks) show at times considerable variation in reliability coefficients. The test–retest reliability of the event trials of the *n*-back tasks, for example, varies between .213 and .603. Fortunately the Monte-Carlo analysis helps us to make sense of these findings. Fig. 1 shows that as data aggregation increases, there are both an increase in the average reliability estimate, and a decrease in the variability in these estimates. Put another way, not only are measures of ISV with few trials unreliable, but estimates of their reliability may be themselves more variable. This suggests that it may be unwise to make comparisons between tasks used here. It further suggests that ISV measures with few trials should be treated with caution, even when reliability estimates seem promising.

Another issue is the decision to remove RTs faster than 120 ms. This precaution against pre-emptive responses is common in standard RT approaches but usually not carried out when using reciprocal methods. Despite this, we decided that comparability across metrics was of cardinal importance, and decided to use the cut-off for all metrics.

Finally, it is unclear to what extent these data can be extrapolated beyond the population of healthy students investigated here. Further work is necessary to explore whether comparable levels of reliability can be found in clinical and developmental populations.

In conclusion, it appears that ISV can be reliably measured across occasions. It also appears that, due to the high redundancy between many of these measures, psychometric reliability should be a key criterion when deciding how to operationalise ISV. These data also make a strong claim for the importance of using sufficient trials when studying ISV, as trial numbers which have proved adequate in research using first moment parameters may prove inadequate when studying the higher moments of the RT distribution.

## Acknowledgements

The authors would like to thank two anonymous reviewers for their helpful comments on the manuscript.

## References

- Birkett, P., Sigmundsson, T., Sharma, T., Touloupoulou, T., Griffiths, T. D., Reveley, A., et al. (2007). Reaction time and sustained attention in schizophrenia and its genetic predisposition. *Schizophrenia Research*, 95, 76–85.
- Brown, W. (1910). Some experimental results in the correlation of mental abilities. *British Journal of Psychology*, 3, 296–322.
- Burton, C. L., Strauss, E., Hultsch, D. F., Moll, A., & Hunter, M. A. (2006). Intraindividual variability as a marker of neurological dysfunction: A comparison of Alzheimer's disease and Parkinson's disease. *Journal of Clinical and Experimental Neuropsychology*, 28(1), 67–83.
- Carpenter, R. H. S. (1981). Oculomotor procrastination. In D. F. Fisher, R. A. Monty, & J. W. Senders (Eds.), *Eye movements: Cognition and visual perception* (pp. 237–246). Hillsdale: Lawrence Erlbaum.
- Carpenter, R. H. S. (1994). SPIC: a PC-based system for rapid measurement of saccadic responses. *Journal of Physiology (Proceedings)*, 480, 4.
- Flehmig, H. C., Steinborn, M., Langner, R., Scholz, A., & Westhoff, K. (2007). Assessing intraindividual variability in sustained attention: reliability, relation to speed and accuracy, and practice effects. *Psychology Science*, 49(2), 132–149.
- Heathcote, A., Brown, S., & Cousineau, D. (2004). QMPE: Estimating Lognormal, Wald, and Weibull RT distributions with a parameter-dependent lower bound. *Behavior Research Methods, Instruments, and Computers*, 36(2), 277–290.
- Hultsch, D. F., MacDonald, S. W. S., & Dixon, R. A. (2002). Variability in reaction time performance of younger and older adults. *Journal of Gerontology: Psychological Sciences*, 57B(2), 101–115.
- Jensen, A. R. (1992). The importance of intraindividual variation in reaction time. *Personality and Individual Differences*, 13(8), 869–881.
- Klein, C., & Fischer, B. (2005). Instrumental and test–retest reliability of saccadic measures. *Biological Psychology*, 68, 201–213.
- Klein, C., Wendling, K., Huettner, P., Ruder, H., & Peper, M. (2006). Intra-subject variability in attention-deficit hyperactivity disorder. *Biological Psychiatry*, 60, 1088–1097.
- Kuntsi, J., Stevenson, J., Oosterlan, J., & Sonuga-Barke, E. J. S. (2001). Test–retest reliability of a new delay aversion task and executive function measures. *British Journal of Developmental Psychology*, 19, 339–348.
- Larson, G. E., & Alderton, D. L. (1990). Reaction time variability and intelligence: A “worst performance” analysis of individual differences. *Intelligence*, 14, 309–325.
- Leth-Steensen, C., King-Elbaz, Z., & Douglas, V. I. (2000). Mean response times, variability, and skew in the responding of ADHD children: A response time distributional approach. *Acta Psychologica*, 104, 167–190.
- MacDonald, S. W. S., Hultsch, D. F., & Dixon, R. A. (2008). Predicting impending death: Inconsistency in speed is a selective and early marker. *Psychology and Aging*, 23(3), 595–607.
- Novick, M. R. (1966). The axioms and principal results of classical test theory. *Journal of Mathematical Psychology*, 3, 1–18.
- Saville, C. W. N., Dean, R. O., Daley, D., Intriligator, J., Boehm, S., Feige, B., et al. (in press). Electrocortical correlates of intra-subject variability in reaction times: Average and single-trial analyses. *Biological Psychology*, doi:10.1016/j.biopsycho.2011.02.005.
- Schmiedek, F., Oberauer, K., Wilhelm, O., Süß, H.-M., & Wittmann, W. W. (2007). Individual differences in components of reaction time distributions and their relations to working memory and intelligence. *Journal of Experimental Psychology: General*, 136(3), 414–429.
- Spearman, C. (1910). Correlation calculated from faulty data. *British Journal of Psychology*, 3, 171–195.